INCH-POUND

MIL-DTL-16377/84
18 December 2008

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, EMERGENCY LIGHTING 54 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 333.1T8EM, 333.2T8EM, 342.2T8EM, AND 342.3T8EM

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

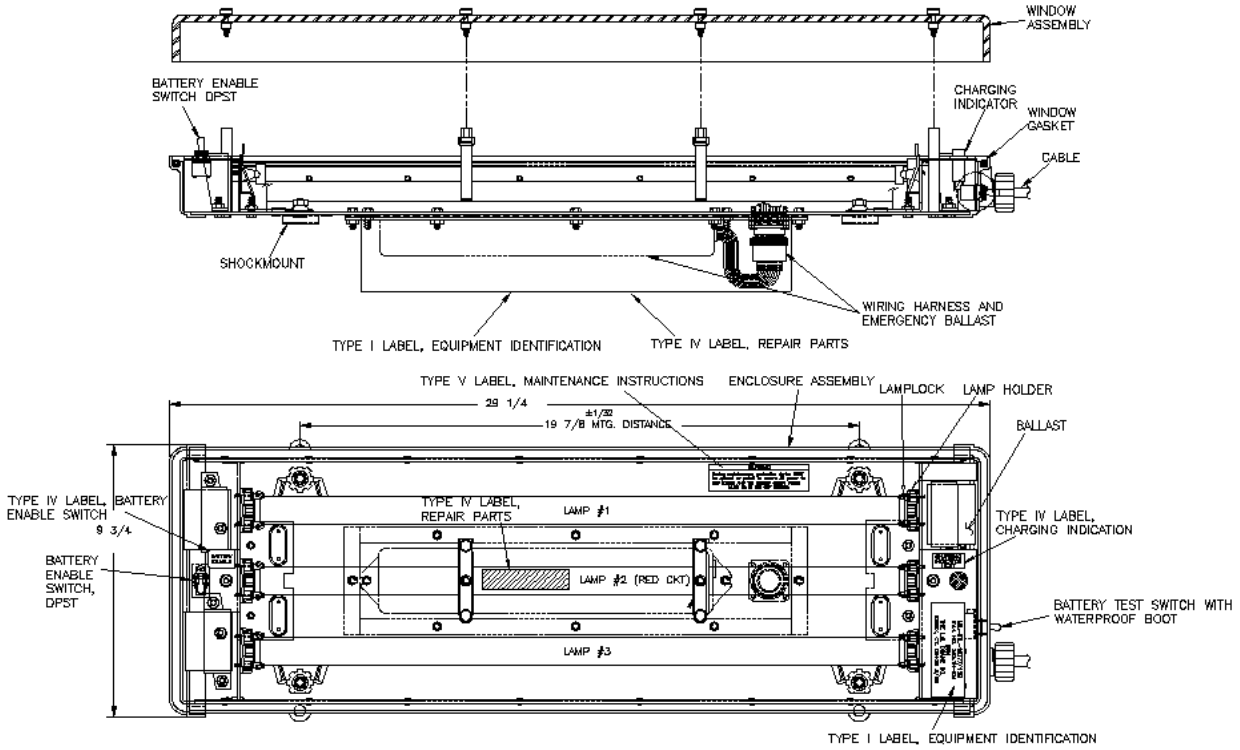


FIGURE 1. Typical view of symbols333.1T8EM and 342.2T8EM, symbols 333.2T8EM, and 342.3T8EM use identical construction with flush mounting flange not shown.

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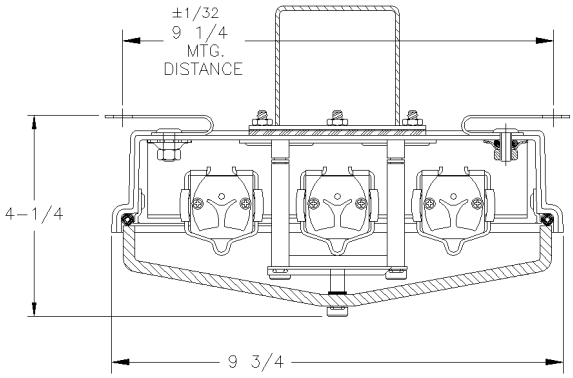
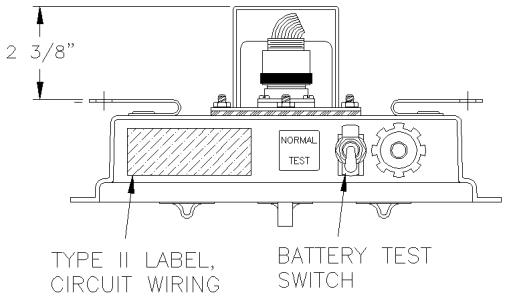


FIGURE 2. Typical section for symbols 333.1T8EM and 342.2T8EM.



NOTES:

1. Symbols 333.2T8EM and 342.3T8EM use identical battery test switch construction with additional switch hardware (not shown) for switch operation below ceiling.
2. Type II label required on all fixtures.

FIGURE 3. Typical section for symbols 333.2T8EM and 342.3T8EM.

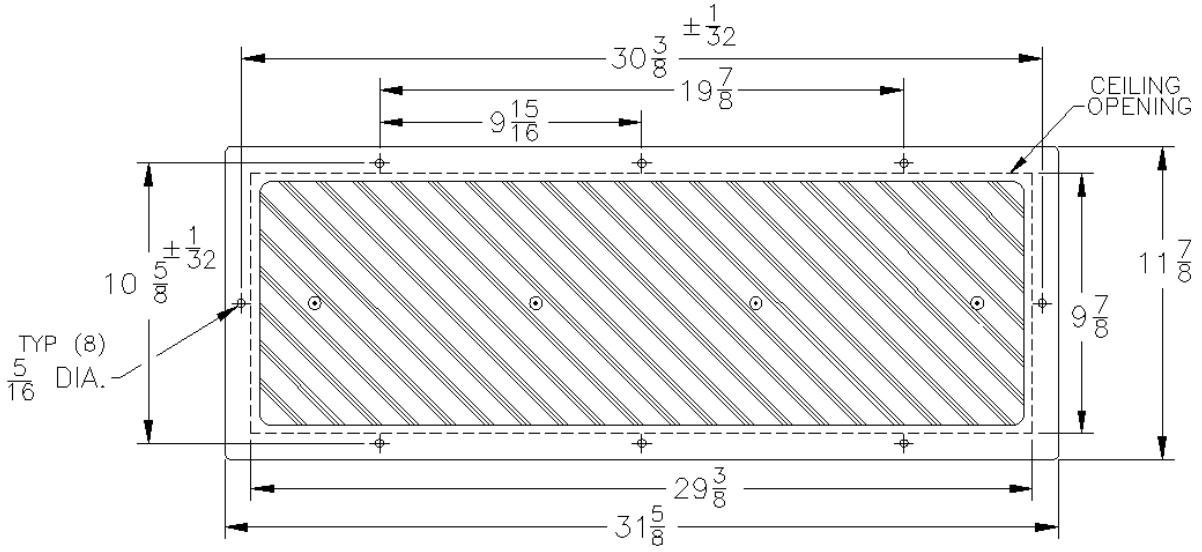


FIGURE 4. Flush mounting dimension for symbols 333.2T8EM and 342.3T8EM.

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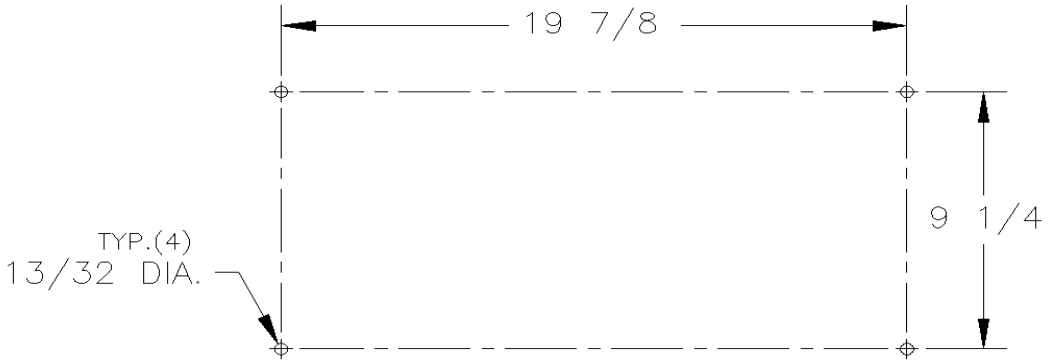
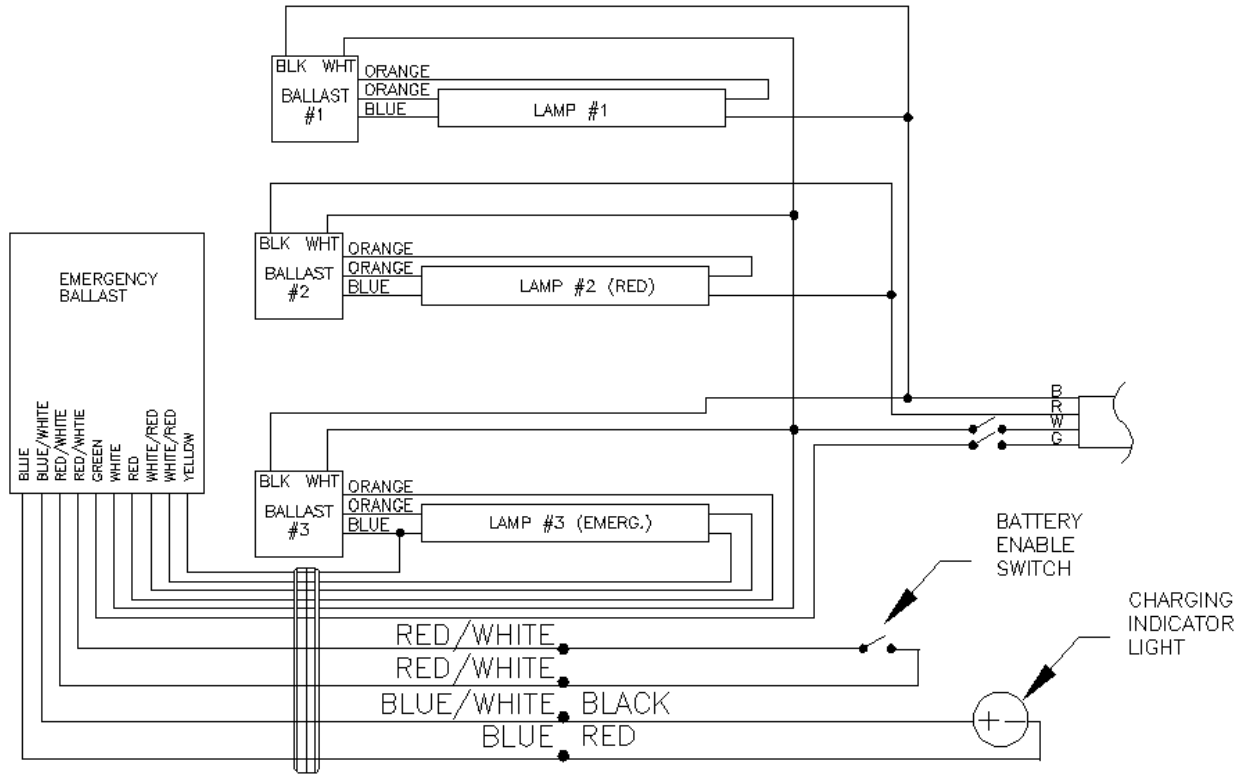


FIGURE 5. Standard mounting dimensions utilizing shock mounts for symbols 333.1T8EM and 342.2T8EM.



NOTES:

1. The black and red input conductors are the only switched power leads to the fixture, used for normal on/off lamp control.
2. The white and green input conductors shall be connected directly to the power source and remain continuously energized for emergency lamp battery charging. The white and green input conductors shall not have power interrupted by on/off control switches.
3. The white input conductor is shared by the normal lamp circuits (white and red) and by the emergency circuit.

FIGURE 6. Typical wiring diagram.

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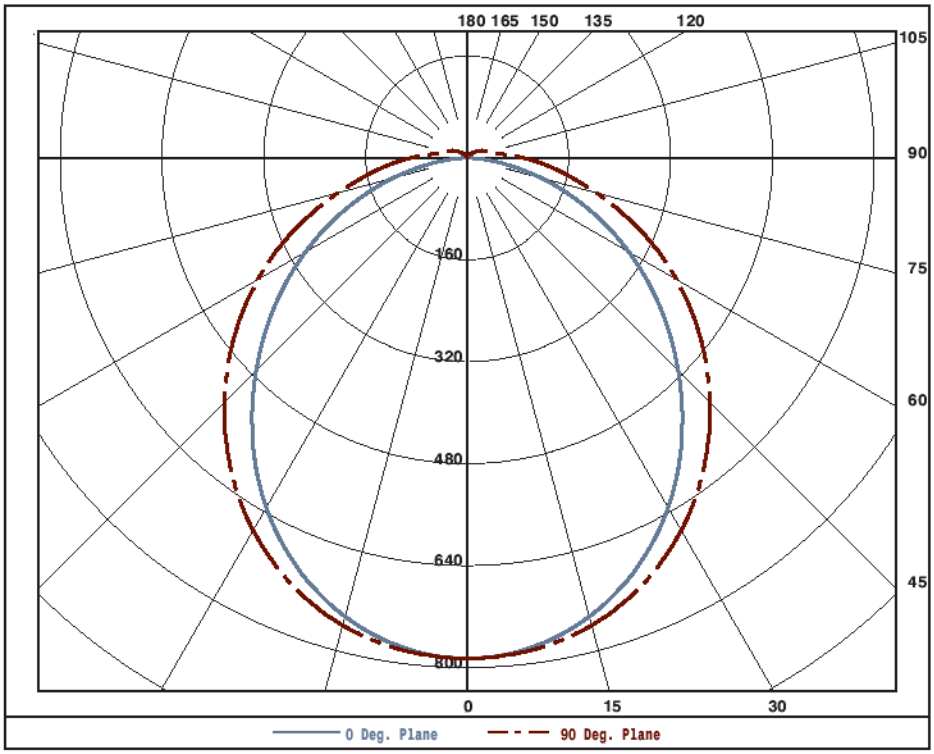


FIGURE 7. Candela distribution curves, white diffusing window, symbols 333.1T8EM and 333.2T8EM.

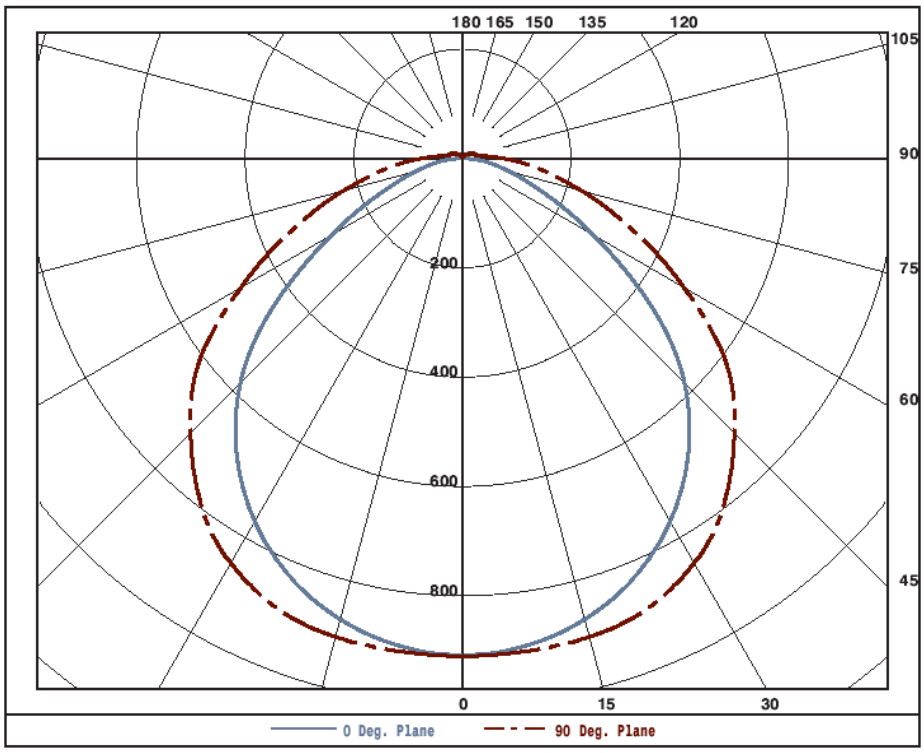


FIGURE 8. Candela distribution curves, prismatic window, symbols 342.2T8EM and 342.3T8EM.

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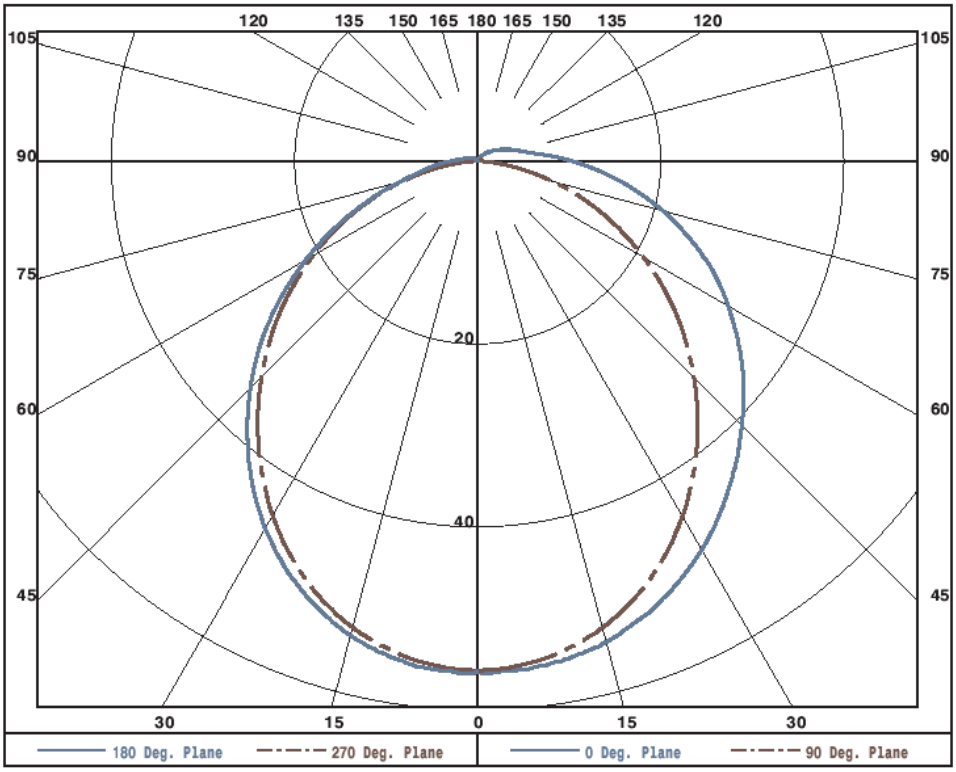


FIGURE 9. Candela distribution curves, emergency mode.

TABLE I. Numerical candela distribution curves, white diffusing window, symbols 333.1T8EM and 333.2T8EM.

Vertical angle	Candela distribution		Vertical angle	Candela distribution	
	0-degree	90-degree		0-degree	90-degree
0	786	786	95	8	62
5	781	783	105	6	39
15	745	760	115	4	26
25	677	709	125	3	17
35	582	634	135	0	10
45	469	540	145	0	3
55	353	436	155	0	0
65	234	323	165	0	0
75	120	211	175	0	0
85	27	124	180	0	0
		90	9	89	

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TABLE II. Numerical candela distribution curves, prismatic window, symbol 342.2T8EM and 342.3T8EM.

Vertical angle	Candela distribution	
	0-degree	90-degree
0	910	910
5	904	910
15	872	906
25	810	881
35	713	817
45	577	709
55	378	566
65	195	381
75	88	227
85	30	119
90	9	76

Vertical angle	Candela distribution	
	0-degree	90-degree
95	8	48
105	6	26
115	5	22
125	3	15
135	1	9
145	0	4
155	0	1
165	0	0
175	0	0
180	0	0

TABLE III. Numerical candela distribution, emergency mode.

Angle	Candlepower	
	Symbol number	
	0-degree	90-degree
0	56	56
5	56	55
15	54	53
25	51	48
35	46	41
45	41	33
55	35	25
65	28	17
75	21	8
85	14	2
90	10	1

Angle	Candlepower	
	Symbol number	
	0-degree	90-degree
95	8	0
105	4	0
115	3	0
125	2	0
135	1	0
145	1	0
155	0	0
165	0	0
175	0	0
180	0	0

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PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other pertinent IESNA procedures.

Plane: zero degrees is parallel to lamps, 90 degrees is perpendicular to lamps.

Electrical values: 120 volts, 60 hertz, 53.7 watts+4.0 watts (battery charging circuit)=57.7 watts total.

Ambient temperature: 25±1 °C.

Lamps: GE F17T8-SP41-ECO (seasoned for at least 100 hours).

Lamps correction: lamps rated at 1325 lumens.

Ballast: Furnished with fixture, see MIL-DTL-16377/81 for 17-watt lamp specification, MIL-DTL-16377/77 for emergency ballast specification.

REQUIREMENTS:

CLASSIFICATION:

Type: I (Fluorescent).

Class: 2 (General).

SYMBOL NUMBERS:

333.1T8EM:	Complete lighting fixture assembly, standard mounting, with three 17-watt cool white fluorescent lamps and white diffusing window assembly, includes emergency battery/ ballast circuit for emergency lighting.
333.2T8EM:	Complete lighting fixture assembly, flush mounting, with three 17-watt cool white fluorescent lamps and white diffusing window assembly, includes emergency battery/ ballast circuit for emergency lighting.
342.2T8EM:	Complete lighting fixture assembly, standard mounting, with three 17-watt cool white fluorescent lamps and clear and white prismatic window assembly, includes emergency battery/ ballast circuit for emergency lighting.
342.3T8EM:	Complete lighting fixture assembly, flush mounting, with three 17-watt cool white fluorescent lamps and clear and white prismatic window assembly, includes emergency battery/ ballast circuit for emergency lighting.

DESIGN:

Electrical rating:	51 watts, 120 volts, 60 hertz, ungrounded electrical distribution system. Total input power: 53.7 watts+4.0 watts (battery charging circuit)=57.7 watts. Total at 120 volts.
Dimension and configurations:	See figures 1, 2, and 3.
Photometric data:	See figures 7, 8, 9, and tables I, II, III.
Wiring diagram:	See figure 6.
Ballast:	See MIL-DTL-16377/81 for 17-watt lamp specification, MIL-DTL-16377/77 for emergency ballast specification.
Mounting dimensions:	
Symbols 333.1T8EM and 342.2T8EM:	Standard mounting (surface), see figure 5.

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Symbols 333.2T8EM and 342.3T8EM:	Flush mounting (recessed), see figures 4.
Window assembly:	
Symbols 333.1T8EM and 333.2T8EM:	White window, see table IV and figure 27 of MIL-DTL-16377.
Symbols 342.2T8EM and 342.3T8EM:	Prismatic window, see table IV and figure 27 of MIL-DTL-16377.
Enclosure:	Watertight; see 3.5.11 of MIL-DTL-16377.
Cable:	10 feet (minimum) of type LSFHOF-3 in accordance with MIL-DTL-24643/3
Weight:	
Symbols 333.1EM and 342.2EM:	Not to exceed 20 pounds.
Symbols 333.2EM and 342.3EM:	Not to exceed 22 pounds.

DESIGNATION AND MARKING:

The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3.

a. Two type I labels, equipment identification, reference MIL-DTL-16377 figure 11, shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.

b. One type II label, circuit wiring information, reference MIL-DTL-16377 figure 12, located on outside surface adjacent to cable entrance.

c. Two type IV label, repair parts, reference MIL-DTL-16377 figure 14, information to be shown shall include "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table IV, located on an inner nonreflecting surface and back of fixture.

d. One type IV label, reference MIL-DTL-16377 figure 14, information to be shown shall be “CHARGING INDICATOR LIGHT”.

e. One type IV label, reference MIL-DTL-16377 figure 14, information to be shown shall be “BATTERY ENABLE”.

f. One type V label, reference MIL-DTL-16377 figure 15, information to be shown shall be “DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION”.

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 as follows:

First article testing/inspection:

Complete fixtures:	See table VII.
Window assemblies:	See table VIII.
Comparison inspection:	See tables VII and VIII.
Conformance inspection:	See MIL-DTL-16377.
Inspection of product for delivery:	
Examination	See MIL-DTL-16377.
Conformance test	See MIL-DTL-16377.

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Examination for preparation of delivery See MIL-DTL-16377.

GENERAL INFORMATION:

Table IV. Fixture standard parts.

Table V. Complete fixture first article tests.

Table VI. Window assembly first article tests.

Table VII. Complete fixture part numbers.

NOTES:

1. See MIL-DTL-16377/42 for T8 color filter tube specifications and part numbers.
2. Installation circuit wiring shall incorporate a minimum ratio of eight 17-watt lamps and one emergency 17-watt lamp. The 8 to 1 ratio is required for meeting MIL-STD-1399-300 harmonic requirements. The 17-watt emergency ballast produces non-linear current that must be offset by the linear current from eight 17-watt lamp ballasts connected to the same circuit.
3. Emergency fixtures may be designated as Navy Emergency Automatic Lighting System or NEALS.
4. The “battery enable” switch shall be placed in the normal position after installation.

TABLE IV. Standard parts.

Name	Military p/n or specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F17T8/CW	6240-01-391-6027	---
Lamplock	M16377/40-003	---	---
Lampholder	UL Standard 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001	---	Integrated starter type
Shockmount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, diffusing	M16377/12-001	6210-00-295-2849	1/
Window assembly, prismatic	M16377/12-003	6210-00-725-9030	1/
Emergency ballast, wiring harnesses, and mating connector	M16377/77-001	6250-01-491-7039	Complete assembly: includes wiring harnesses and matting AMP #206043 connectors.
Charging indicator light	---	6220-01-415-7889	LED type
Switch, battery enable	M16377/75-008	5930-01-414-9240	DPST
Switch, battery test	Carling 2167-M-A-NC-73XCA4	---	Momentary toggle
Switch cover, rubber boot	MIL-DTL-5423/2	---	Neoprene

NOTE:
1/ Window assemblies (diffusing and prismatic) are the same as T12 lamp fixtures (MIL-DTL-16377/12).

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TABLE V. First article tests and comparison inspection for complete fixtures.[1/]

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Electromagnetic Interference	3.6.14	4.8.16	---
Electrical equipment interface testing requirements: MIL-STD-1399-300	MIL-STD-1399-300	MIL-STD-1399-300	[2/], [3/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/]
Shock	3.5.12	4.8.7	[4/]
Vibration	3.5.13	4.8.8	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Noise	3.5.14	4.8.9	---
Continuity of grounding	3.6.13.4	4.8.12	---
Light output	3.7.1	4.8.6.1	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Emergency light output: Distribution curves shall be documented during emergency lamp operation	3.7.1	4.8.6.1	[5/]
Emergency ballast performance: Operating time and discharge test	See MIL-DTL-16377/77, Table II, battery discharge test	See MIL-DTL-16377/77, Table II, battery discharge test	---

NOTES:

1/ Requirements and test methods use MIL-DTL-16377.

2/ If less, then 1KVA exception not applicable for input waveform THD.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes, (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300.

4/ Tests shall be performed in the order specified.

5/ Shall meet or exceed published distribution curves.

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TABLE VI. First article tests and comparison inspection for window assemblies.[1/]

Test	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/]
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
[1/] Testing requirements and methods use MIL-DTL-16377.
[2/] Tests shall be performed in the order specified.

TABLE VII. Complete fixture part numbers.

Name	Military p/n
Complete fixture, Symbol 333.1T8EM	M16377/84-333.1T8EM
Complete fixture, Symbol 333.2T8EM	M16377/84-333.2T8EM
Complete fixture, Symbol 342.2T8EM	M16377/84-342.2T8EM
Complete fixture, Symbol 343.3T8EM	M16377/84-342.3T8EM

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Preparing Activity:
Navy – SH
(Project 6210-2006-008)

Review Activities:
Navy – AS, CG
Air Force – 03
DLA – GS2
GSA – FAS

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